Correlation

PS50008A Week 15 | Lecture

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Learning Outcomes

By the end of this lecture you will be able to:

- Describe relationships between continuous variables
- Create and interpret scatterplots
- Calculate and interpret Pearson's r and Spearman's rho
- Explain why correlation does not imply causation

Relationships Between Variables

Beyond Group Differences

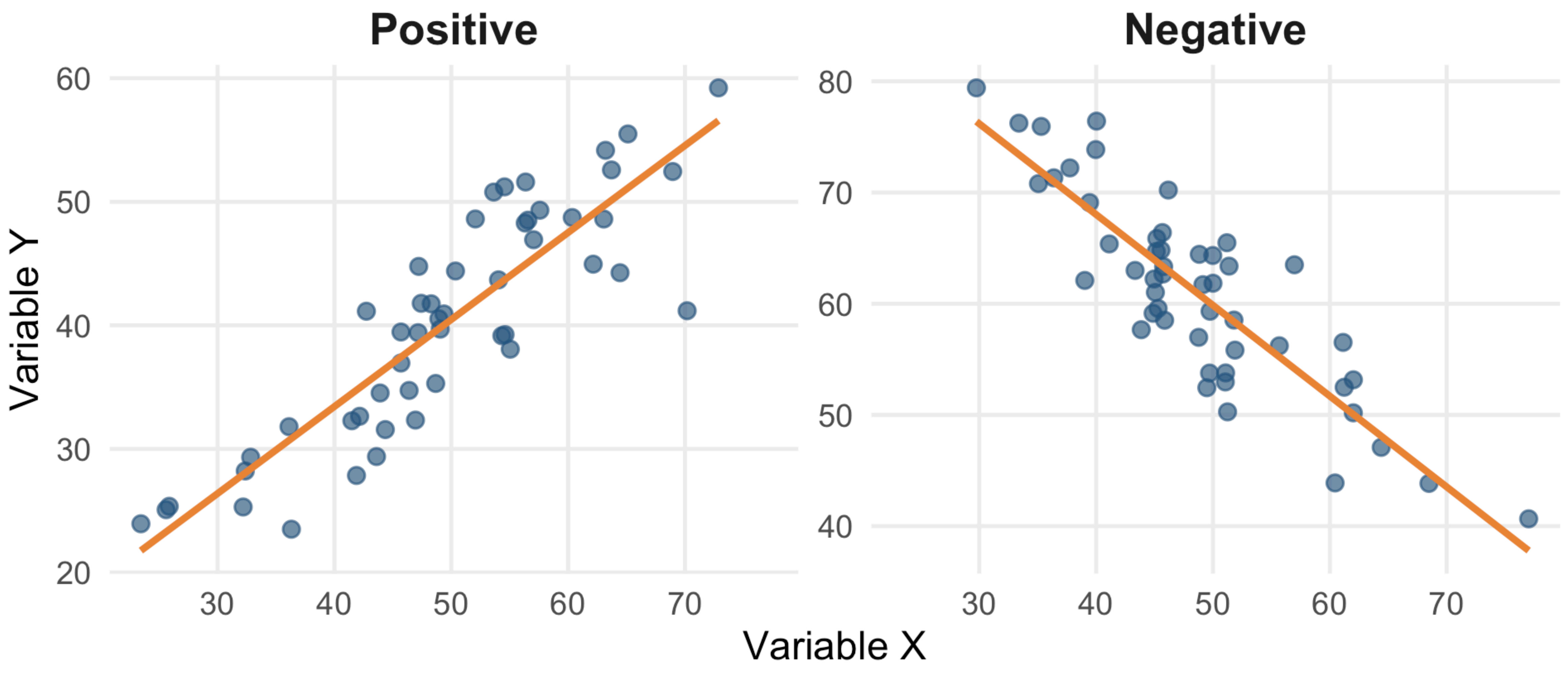
- So far: comparing groups (t-tests)
- Now: examining relationships between variables
- Both variables are measured, not manipulated

> **Question:** Is there a relationship between study hours and exam scores?

What is Correlation?

> **Correlation**: A statistical measure that quantifies the strength and direction of the linear relationship between two continuous variables.

Direction of Relationships



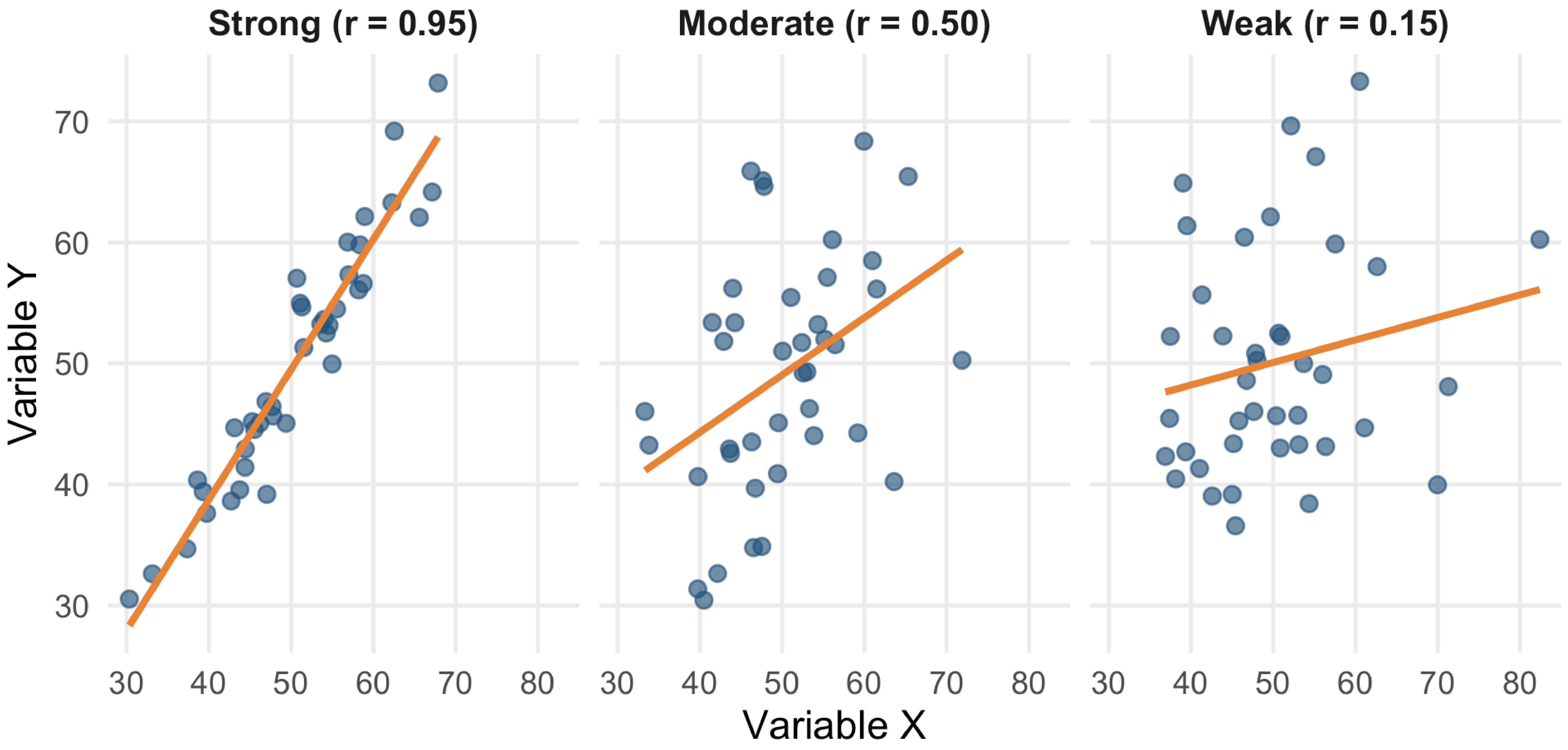
Positive: As X ↑, Y ↑

Example: Height and weight

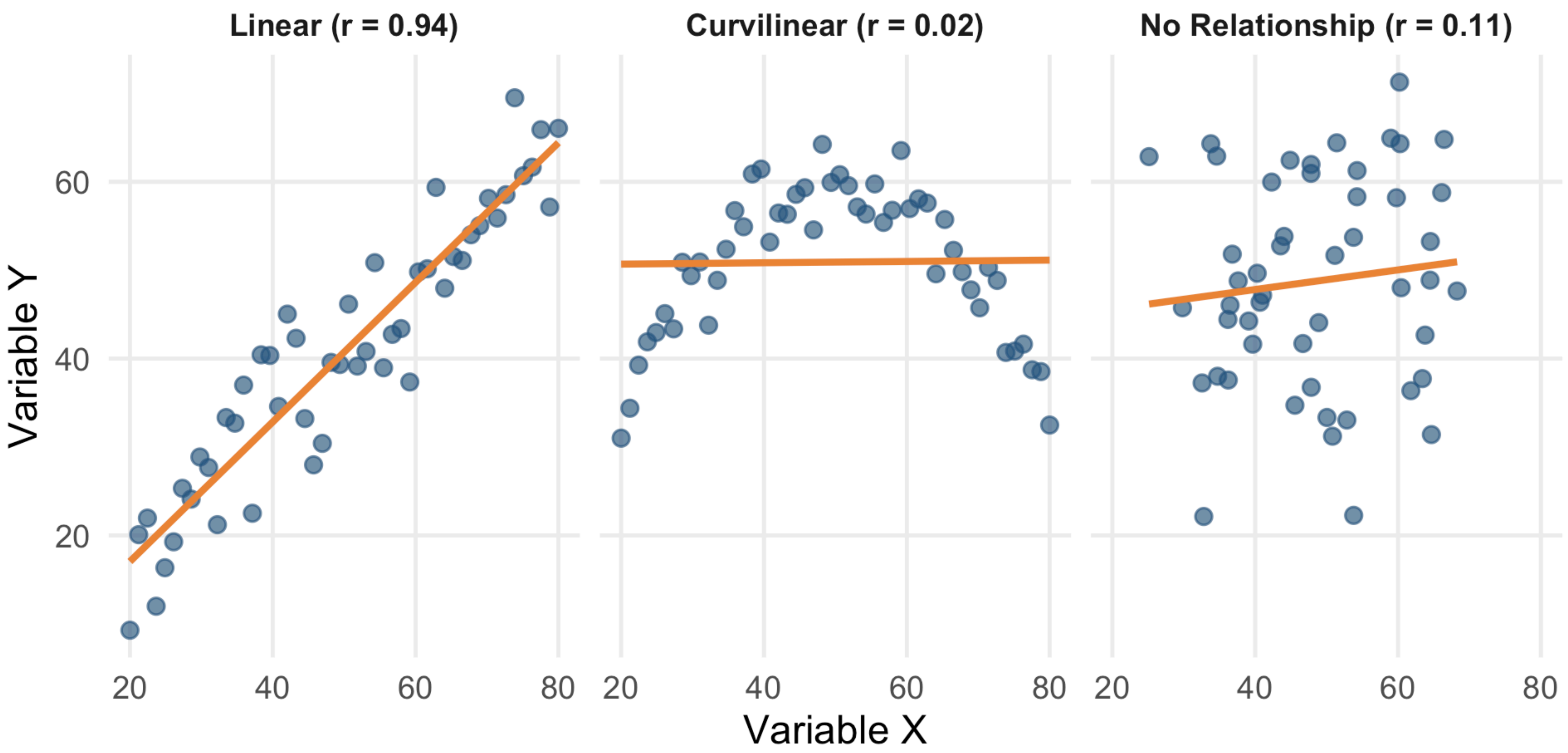
Negative: As X ↑, Y ↓

Example: Stress and sleep quality

Strength of Relationships



Types of Relationships



> ⚠ **Warning**
>
> Correlation measures LINEAR relationships only. The curvilinear data has a strong pattern but r ≈ 0!

Scatterplots

Visualising Relationships

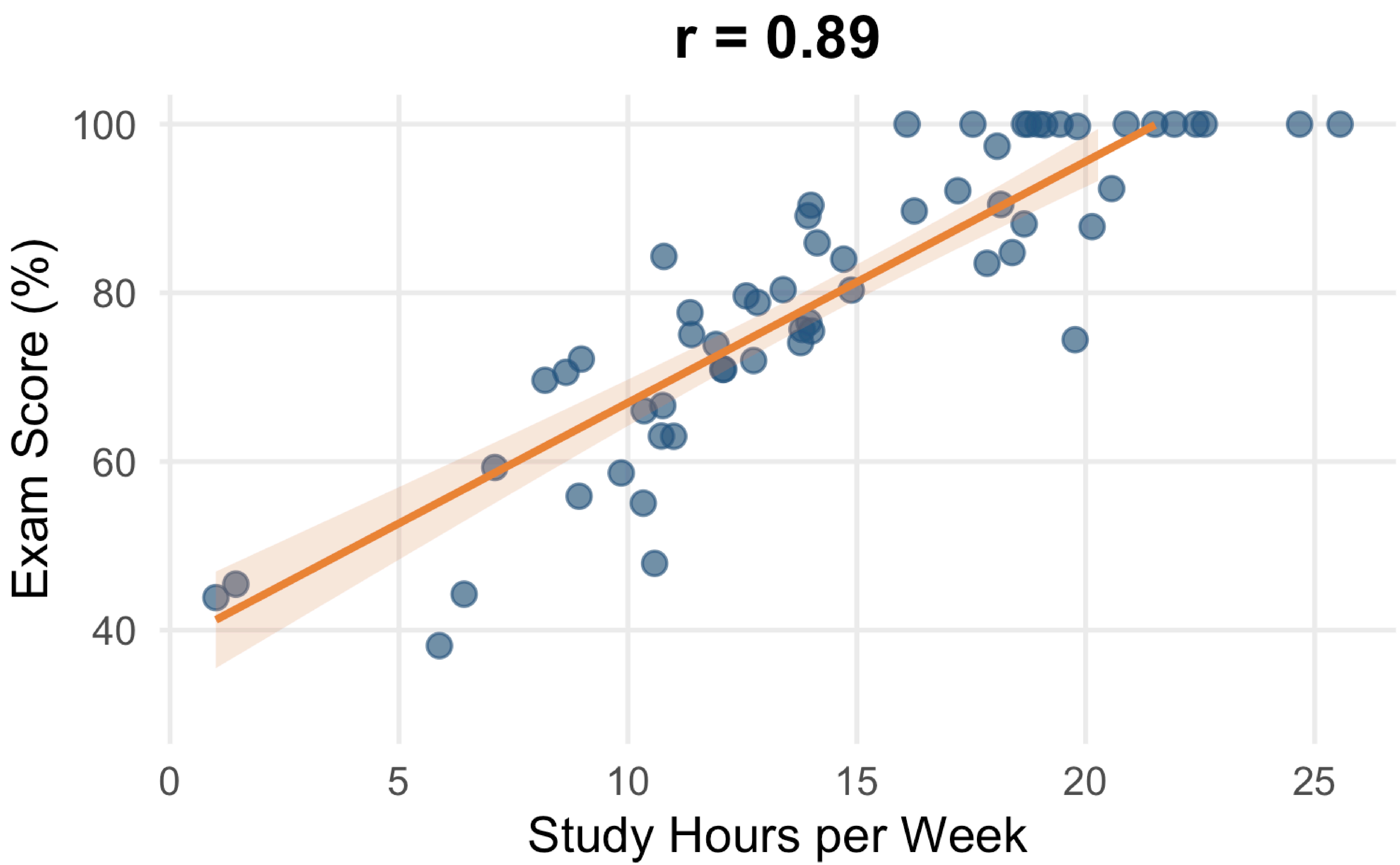
> **Scatterplot**: A graph displaying the relationship between two continuous variables, with each observation represented as a point.

- X-axis: One variable (often predictor)
- Y-axis: Other variable (often outcome)
- Each point: One observation

Creating Good Scatterplots

- Label axes clearly (with units)
- Use appropriate scales
- Include a trend line (with caution)
- Look for patterns, outliers, clusters

Example: Study Hours vs Exam Scores

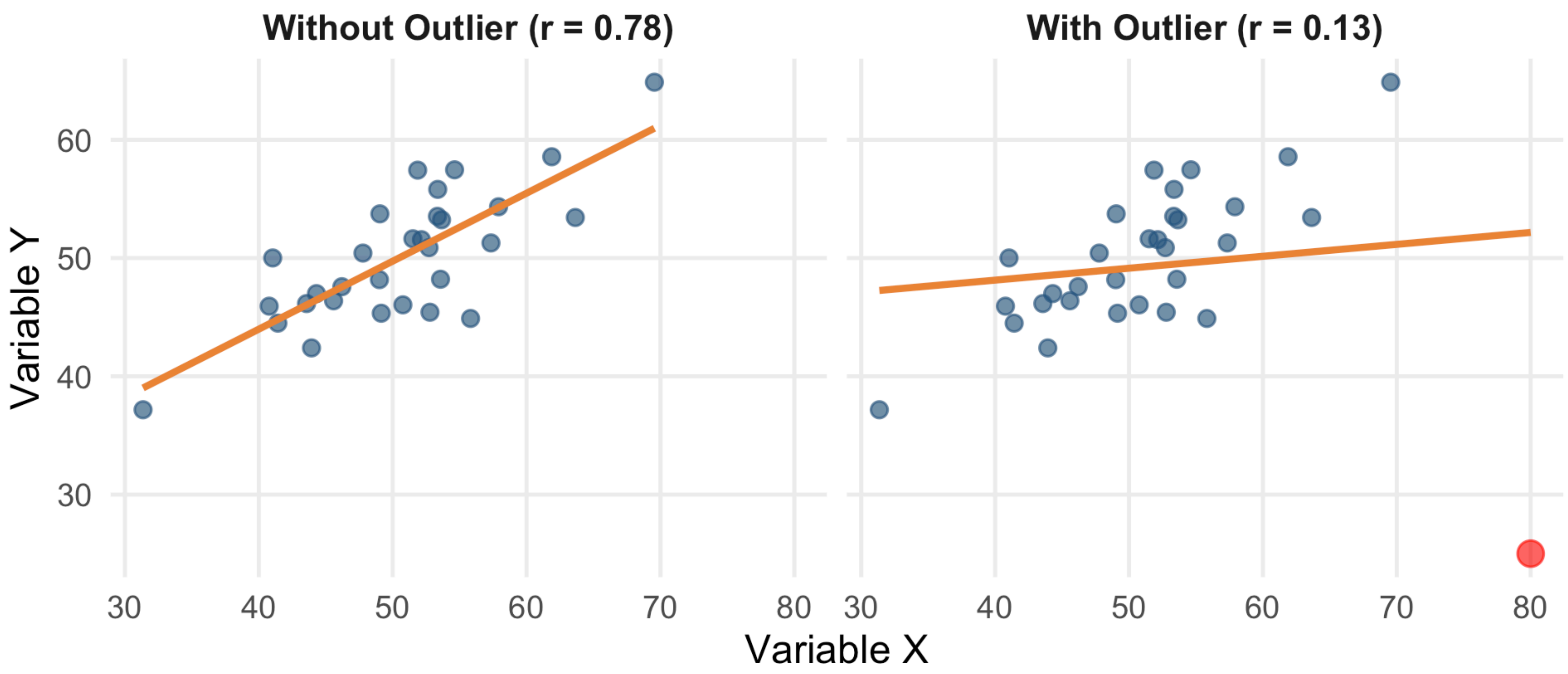


What to Look For

When examining a scatterplot, ask:

1. Is there a relationship? (or random scatter?)
2. What direction? (positive/negative)
3. How strong? (tight/loose clustering)
4. Is it linear? (or curved?)
5. Any outliers? (unusual points)

The Impact of Outliers



⚠ **Warning**

A single outlier dramatically changed r from 0.70 to 0.35! Always visualise first.

Pearson’s r and Spearman’s rho

Pearson's Correlation Coefficient

Pearson's r: A measure of the linear relationship between two continuous variables. Ranges from -1 to +1.

$$r = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{(n-1) s_X s_Y}$$

Interpreting Pearson’s r

r value	Interpretation
+/- 1.0	Perfect correlation
+/- 0.7 to 0.9	Strong
+/- 0.4 to 0.6	Moderate
+/- 0.1 to 0.3	Weak
0	No linear relationship

Note

These are rough guidelines. Context and field norms matter!

Coefficient of Determination (r^2)

r^2 (r-squared): The proportion of variance in Y that is explained by X. Found by squaring r.

If $r = 0.6$, then $r^2 = 0.36$

“36% of the variance in exam scores is explained by study hours.”

Assumptions of Pearson’s r

- Both variables are continuous
- Linear relationship (check scatterplot!)
- Bivariate normality (roughly)
- No extreme outliers
- Independence of observations

When Pearson's r Fails

⚠ **Warning**

Pearson's r is inappropriate when:

- Data are ordinal (ranked)
- Relationship is non-linear
- There are extreme outliers
- Data are severely non-normal

Spearman’s Rho (rho)

> **Spearman’s rho**: A non-parametric correlation based on ranks rather than raw values. More robust to outliers and non-linearity.

- Converts data to ranks
- Then calculates Pearson’s r on ranks
- Ranges from -1 to +1 (same interpretation)

Pearson vs Spearman

Use Pearson's r when:

- Both variables continuous
- Linear relationship
- Roughly normal data
- No major outliers

Use Spearman's rho when:

- Ordinal data
- Non-linear but monotonic
- Non-normal data
- Outliers present

Testing Significance

Null hypothesis: $r = 0$ (no correlation in population)

Test: Is our observed r significantly different from 0?

Report: $r(df) = \text{value}$, $p = \text{value}$

Correlation Does Not Equal Causation

The Third Variable Problem

Third Variable (Confound): An unmeasured variable that causes both X and Y, creating a spurious correlation between them.

Ice cream sales and drowning deaths are correlated.

Third variable: Hot weather causes both!

Directionality Problem

- Correlation shows X and Y are related
- But which causes which?
- X -> Y?
- Y -> X?
- Or both directions?

> **Depression and social isolation are correlated.**
>
> Does depression cause isolation, or isolation cause depression, or both?

Famous Spurious Correlations

Real correlations that aren't causal:

- Number of pirates and global temperature ($r = -0.9$)
- Nicolas Cage films and pool drownings ($r = 0.87$)
- Cheese consumption and death by bedsheet ($r = 0.95$)

tylervigen.com/spurious-correlations

When CAN We Infer Causation?

- **Temporal precedence**: Cause before effect
- **Covariation**: They vary together
- **Elimination of alternatives**: No confounds

Tip

Experimental designs (with random assignment) can establish causation. Correlational designs cannot.

Reporting Correlations

APA format:

"There was a strong positive correlation between study hours and exam scores, $r(48) = .72$, $p < .001$. Study time explained 52% of the variance in exam performance."

Summary

Key Takeaways

- **Correlation** measures strength and direction of linear relationships
- **Scatterplots** visualise relationships - always look first!
- **Pearson's r** for continuous, linear, normal data
- **Spearman's rho** for ordinal, non-linear, or non-normal data
- **Correlation NEVER proves causation** (third variables, directionality)

Next Week

Reading Week

Questions?

Questions?

See you in the lab!